

Text Mining Final Project Proposal

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A. Introduction

Due to the convenience of technology nowadays, users can comment and express their opinions and experiences on the Internet effortlessly. These users' comments and feedback significantly influence not only on the social platforms but also on business operations. Therefore, it is important to analyze users' comments to uncover valuable insights and to understand user sentiments, which can inform strategic decisions for companies.

In this final project, we propose to use Amazon Books Reviews as a case study to extract and interpret the sentiments expressed by users. By implementing text mining and natural language processing techniques, our primary goal is to acquire keywords and sentiments specific to different genres of books.

B. Data Description

The data comes from Kaggle, which contains 2 csv files. The first is about books reviews that have more than 3 million users' feedback, the second is regarding books details with over 21 thousand unique books.

C. Methods (Test/Evaluation)

For the process of analysis, we intend to utilize several python packages to reach our goals, e.g., nltk, sklearn, TextBlob, spaCy, genism, and BERTopic. Figure 1 presents the details of processing.

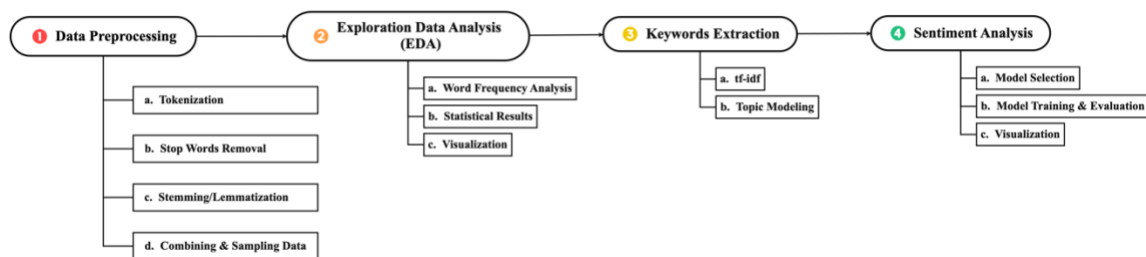


Figure 1. The Flow Chart of Text Mining Project

D. Expected Results

Through this final project, we anticipate that

Question 1: “What are customer complaining about?”

Question 2: “What are the features of popular books?”

could be answered. In addition, the constructed classifiers might be used for future sentiment research to discover more hidden knowledge.